

(1001CJA102120006)

Test Pattern



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

07-03-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B	A,C,D	B	A	A,B,C,D	A,B,D	B	A	A	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	1.49 to 1.50	0.33	28.26 to 28.29	0.18	100.00	1.00 to 1.01	7.85 to 7.86	0.42 to 0.43		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	C,D	A,B,C	A,B,C	A,B,C,D	A,B,C,D	A	C	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	0.20	0.40	12.50	48.80	54.44	1.50	12.00	4.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A,C,D	C	A,C,D	A,C,D	A,B,D	D	C	C	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	3.00	5.65 or 5.66	4.00	18.84 or 18.85	4.24	8.00	5.00	0.11		

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ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,D	A,B,D	A,B,C	C	C,D	A,C	D	C	A	B
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	20.89 to 20.92	1.01 to 1.02	9.00	0.55 to 0.56	9.50 to 9.70	44.00 to 44.10	0.00	1.00		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C,D	A,B,C	A,D	A,D	A,B,D	A,B	A	C	A	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	3.00	6.00	9.00	3.00	6.00	4.00	80.00	6.25		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C,D	B,C	D	C	A,B,C,D	B	B	C	A	D
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	0.75	4.00	3.57	6.00	8.00	1.00	1.00	1.00		



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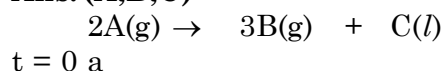
PAPER-1

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (A,B,C)



t = 0 a

$$t = t_a - x \quad \frac{3x}{2} \quad \frac{x}{2}$$

$$t = \infty a - a \quad \frac{3a}{2} + v.p$$

$$\therefore \frac{3a}{2} + v.p = 0.8;$$

and a = 0.2 bar

and v.p. = 0.8 bar

$$t = 10 \text{ min } a - x + \frac{3x}{2} + \frac{x}{2} = 0.3$$

$$\Rightarrow x = 0.1$$

$$\frac{-1}{2} \frac{d[A]}{dt} = k[A]$$

$$\ln \left(\frac{[A]_0}{[A]_t} \right) = 2kt$$

$$\Rightarrow K = \left(\frac{\ln 2}{20} \right) \text{min}^{-1}$$

2. Ans. (C,D)

$$\Delta T_b = 5.4 = 1.8 \times \frac{0.4}{M}$$

$$\Rightarrow M = \frac{4}{30} \text{ kg} = \frac{4000}{30} \text{ g} = \frac{400}{3} \text{ gm}$$

$$\Rightarrow \text{ice separated} = \left(500 - \frac{400}{3} \right) \text{ g} = \frac{1100}{3} \text{ g}$$

$$\Rightarrow X_{\text{glucose}} = \frac{0.4}{\left(\frac{400}{3 \times 18} + 0.4 \right)} = \frac{27}{527}$$

3. Ans. (A,B,C)

4. Ans. (A,B,C)

5. Ans. (A,B,C,D)

6. Ans. (A,B,C,D)

7. Ans.(A)

8. Ans.(C)

9. Ans.(B)

10. Ans.(C)

SECTION-II

1. Ans. (0.20)

$$M = \frac{+2}{0.6} = x(4) + (1-x) + (+3)$$

$$\Rightarrow 4x - 3x + 3 = \frac{2}{0.6}$$

$$\Rightarrow x = \frac{2-1.8}{0.6} = \frac{1}{3}$$

 $\Rightarrow$ frⁿ of atoms in +4 oxidation state = 1/3frⁿ of atoms in +3 oxidation state = 2/3

$$x = \frac{2-1.8}{2} \times 100 = 10\%$$

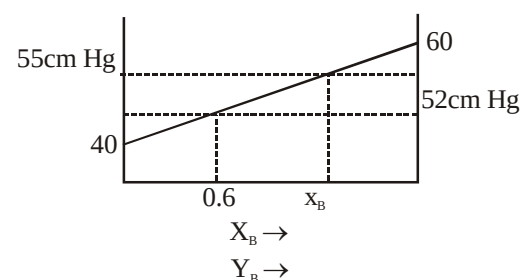
$$y = \left(\frac{2}{1} \right) = 2$$

$$\Rightarrow y/x = 0.2$$

2. Ans. (0.40)

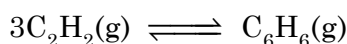
$$X_A = \frac{4}{10} = 0.4 ; X_B = 0.6$$

$$P_s = P_A^0(0.4) + P_B^0(0.6) \\ = 16 + 32 = 52$$



at 55 cm Hg, the liquid composition would be $x_A = 0.4 ; x_B = 0.6$

3. Ans. (12.50)



$$\left(\frac{1-\alpha}{1-\frac{2\alpha}{3}} \right) \quad \left(\frac{\frac{\alpha}{3}}{1-\frac{2\alpha}{3}} \right)$$

$$\Rightarrow \alpha = 0.6$$

$$\Rightarrow \left(\frac{2}{3} \right) \times 0.3 \quad \left(\frac{1}{3} \right) \times 0.3$$

$$\Rightarrow 0.2 \text{ atm} \quad 0.1 \text{ atm}$$

$$\Rightarrow k_p = \frac{0.1}{(0.2)^3} = 12.5 \text{ atm}^{-2}$$

4. Ans. (48.80)

$$\ln \left(\frac{a}{0.8a} \right) = K(20 \text{ min})$$

$$\ln \left(\frac{a_0}{a_t} \right) = K(60 \text{ min})$$

$$\Rightarrow \frac{\ln \left(\frac{a_0}{a_t} \right)}{\ln(1.25)} = 3$$

$$\Rightarrow \frac{a_0}{a_t} = 1.95$$

$$\Rightarrow \% \text{ completion} = 48.8 \%$$

5. Ans. (54.44)

6. Ans. (1.50)

7. Ans. (12.00)

8. Ans. (4.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (C)

Sol. $\lim_{x \rightarrow -\infty} \sqrt{x^2 - \frac{4x}{3}} + x + 2$

$$\lim_{x \rightarrow -\infty} \left(\frac{x^2 - \frac{4x}{3} - (x+2)^2}{\sqrt{x^2 - \frac{4x}{3}} - (x+2)} \right)$$

$$\lim_{x \rightarrow -\infty} \frac{-\frac{16}{3}x - 4}{|x| \sqrt{1 - \frac{4}{3x}} - (x+2)}$$

$$\lim_{x \rightarrow -\infty} \frac{x \left(-\frac{16}{3} - \frac{4}{x} \right)}{x \left[-\sqrt{1 - \frac{4}{3x}} - \left(1 + \frac{2}{x} \right) \right]}$$

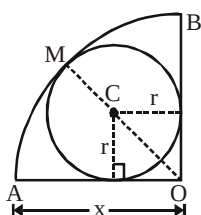
$$\lim_{x \rightarrow -\infty} -\frac{16}{3(-2)} = \frac{8}{3}$$

2. Ans. (A,C,D)

Sol. OM = OC + CM

$$\Rightarrow x = r\sqrt{2} + r$$

$$\Rightarrow \frac{x}{r} = \sqrt{2} + 1 \Rightarrow \frac{r}{x} = \sqrt{2} - 1$$



$$\Rightarrow \frac{x}{r} = \sqrt{2} + 1 \Rightarrow \frac{r}{x} = \sqrt{2} - 1$$

$$\Rightarrow \frac{dx}{dr} = \sqrt{2} + 1 \text{ and } \frac{dr}{dx} = \sqrt{2} - 1$$

$$\text{and } \frac{d^2x}{dr^2} = 0.$$

Now

$$\left(\frac{dr}{dx} \right)^2 + \left(\frac{dx}{dr} \right)^2 = (\sqrt{2} + 1)^2 + (\sqrt{2} - 1)^2 = 2(2+1) = 6$$

$$\& \frac{x+r}{x-r} = \frac{\sqrt{2}+2}{\sqrt{2}} \text{ and } \frac{r-x}{r+x} = \frac{\sqrt{2}-2}{\sqrt{2}}$$

$$\Rightarrow \frac{x+r}{x-r} + \frac{r-x}{r+x} = 2$$

3. Ans. (C)

Sol. $f(x) + f(y) = f(x)f(y) + f(xy)$

partial differentiation wrt x

$$f'(x) = f'(x)f(y) + f'(xy) \cdot (y)$$

put x = 1

$$-2 = -2f(y) + f'(y)y$$

$$\text{get } f(y) = 1 - y^2 \Rightarrow f(x) = 1 - x^2$$

4. Ans. (A,C,D)

Sol. $f(|x|) = \{ |x| - 2, 0 \leq |x| \leq 3 \}$

$$\Rightarrow f(|x|) = \begin{cases} -x-2, & -3 \leq x \leq 0 \\ x-2, & 0 < x \leq 3 \end{cases}$$

$$\Rightarrow [f(|x|)] = \begin{cases} [-x] - 2, & -3 \leq x \leq 0 \\ [x] - 2, & 0 < x \leq 3 \end{cases}$$

$$\text{Also } |f(x)| = \begin{cases} 2, & -3 \leq x \leq 0 \\ 2-x, & 0 < x \leq 2 \\ x-2, & 2 < x \leq 3 \end{cases}$$

$$\Rightarrow [f(x)] = \begin{cases} 2, & -3 \leq x \leq 0 \\ 2+[-x], & 0 < x \leq 2 \\ [x] - 2, & 2 < x \leq 3 \end{cases}$$

So

$$[f(|x|)] + [f(x)] = \begin{cases} [-x], & -3 \leq x \leq 0 \\ [x] + [-x], & 0 < x \leq 2 \\ 2[x] - 4, & 2 < x \leq 3 \end{cases}$$

Now, options can be checked directly without further simplification.

5. **Ans. (A, C, D)**

$$\text{Sol. } f'(x) = \frac{2x^2}{1+x^4} - \frac{1}{2}$$

$$g(x) = \frac{2x^2}{1+x^4}$$

$$g'(x) = \frac{-4x(x-1)(x+1)}{(1+x^4)^2}$$

6. **Ans. (A, B, D)**

$$\text{Sol. } f(x) = (x-1)\tan^{-1}x - \frac{1}{2}\ln(1+x^2)$$

$$f'(x) = \tan^{-1}x - \frac{1}{1+x^2}$$

$$f'(x) > 0 \quad \forall x \geq 1$$

$\Rightarrow f(x)$ is increasing function

$$f''(x) > 0$$

$\Rightarrow f'(x)$ is increasing function $\forall x \geq 1$

$$\lim_{x \rightarrow \infty} f'(x) \rightarrow \frac{\pi}{2} \Rightarrow f'(x) < \frac{\pi}{2} \quad \forall x \geq 1$$

Using LMVT in $[x, x+4]$, $x \geq 1$

$$\frac{f(x+4) - f(x)}{x+4-x} = f'(c) \text{ for } c \in (x, x+4)$$

$$\therefore f(x+4) - f(x) < 2\pi \quad \forall x \in [1, \infty)$$

Paragraph for Question 7 to 8

7. **Ans. (D)**

$$\text{Sol. } \alpha + \beta > 0 \quad \alpha\beta > 0 \quad D \geq 0$$

8. **Ans. (C)**

$$\text{Sol. } \alpha + 2\alpha = 2(1+2k) \quad \alpha 2\alpha = 8+k$$

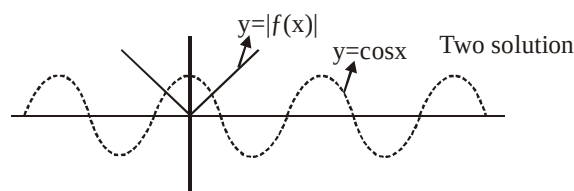
$$2\left(\frac{2(1+2k)}{3}\right)^2 = 8+ka$$

$$32k^2 + 23k = 64$$

Paragraph for Question 9 & 10

9. **Ans. (C)**

$$\text{Sol. } f(x) = 2x$$



10. **Ans. (A)**

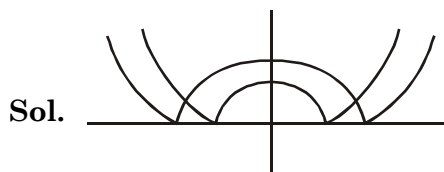
$$\text{Sol. } h(x) = |(2x)(e^{2x} - 1)(2x - 1)(2x - 2)| \text{ is not differentiable at } x = \frac{1}{2} \text{ \& } x = 1.$$

SECTION-II

1. **Ans. 3.00**

$$\text{Sol. } 2x^3 + 5x^2 + 2x - 1 = (x+1)(2x^2 + 3x - 1) \\ \Rightarrow \text{either } x = -1 \text{ is root of equation} \\ \text{or } x^2 + ax + b = 0 \text{ \& } 2x^2 + 3x - 1 = 0 \text{ have both roots in common.}$$

2. **Ans. (5.65 or 5.66)**



Sol.

Point of intersection when

$$x^2 - 1 = 3 - x^2 \Rightarrow x = \sqrt{2}$$

$$y = x^2 - 1 \text{ \& } y = 3 - x^2$$

$$\left(\frac{dy}{dx}\right)_1 = 2\sqrt{2} \text{ \& } \left(\frac{dy}{dx}\right)_2 = -2\sqrt{2}$$

$$\tan \theta = \frac{4\sqrt{2}}{1-8} \Rightarrow \theta = \tan^{-1}\left(\frac{4\sqrt{2}}{7}\right)$$

3. **Ans. 4.00**

$$\text{Sol. } P(x) = Q(x)(x-3) + 6$$

Similarly $x = 3$

$$P(3) = 6$$

$$P(-3) = -6$$

$$P(x) = Q'(x)(x^2 - 9) + ax + b$$

$$3a + b = P(3) = 6$$

$$-3a + b = P(-3) = -6$$

$$a = 2 \quad b = 0$$

$$g(x) = 2x$$

$$g(2) = 4$$

4. **Ans. 18.84 or 18.85**

Sol. $\frac{dy}{dx} = \cos(x-y) \left[1 - \frac{dy}{dx} \right]$

$$\frac{dy}{dx} = \frac{\cos(x-y)}{1 + \cos(x-y)} = \frac{1}{2}$$

$$\therefore \cos(x-y) = 1$$

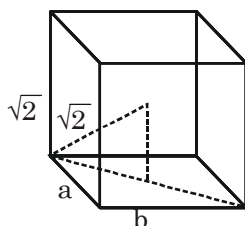
$$\Rightarrow \sin(x-y) = 0$$

$$\Rightarrow y = 0$$

$$\therefore \cos x = 1 \Rightarrow x = -4\pi, -2\pi, 0, 2\pi, 4\pi, 6\pi$$

5. **Ans. 4.24**

Sol. Let the other edges be 'a' & 'b'



$$\left(\frac{\sqrt{a^2 + b^2}}{2} \right)^2 + \left(\frac{\sqrt{2}}{2} \right)^2 = 2$$

$$\frac{a^2 + b^2}{4} = \frac{3}{2} \Rightarrow a^2 + b^2 = 6$$

Now, volume of cuboid = $ab\sqrt{2}$

$$\therefore \frac{a^2 + b^2}{2} \geq \sqrt{a^2 b^2}$$

$$\frac{a^2 + b^2}{2} \geq ab \Rightarrow ab \leq 3$$

$$\therefore V_{\max} = 3\sqrt{2}$$

6. **Ans. 8.00**

Sol. $\lim_{x \rightarrow 0} \left(\frac{f(x) - x + 2x^2 - 3x^3}{3x^3} \right) \cdot 2 = 4$

$$\Rightarrow f(x) = x - 2x^2 + 3x^3 + 6x^4$$

$$\Rightarrow \sum_{i=0}^4 a_i = 1 - 2 + 3 + 6 = 8$$

7. **Ans. 5.00**

Sol. $\frac{f'(x)}{f(x)} = \frac{{}^nC_0}{{}^nC_0 x + 1} + \frac{{}^nC_1}{{}^nC_1 x + 2} + \dots + \frac{{}^nC_n}{{}^nC_n x + n + 1}$

$$\Rightarrow \frac{f'(0)}{f(0)} = \frac{{}^nC_0}{1} + \frac{{}^nC_1}{2} + \dots + \frac{{}^nC_n}{n+1}$$

$$f'(0) = (n+1)! \left(\frac{2^{n+1} - 1}{n+1} \right) = n!(2^{n+1} - 1)$$

$$\therefore n!(2^{n+1} - 1) = \frac{3}{2} 7!$$

$$\Rightarrow n = 5$$

8. **Ans. 0.11**

Sol. $x = 1, f(1) + 2f'(1) = 1$

$$\Rightarrow f(1) = \frac{1}{3}$$

$$f(x^2) + 2x^2 f'(x^2) + (x^2 + 1)f'(x) + 2xf'(x) = 1$$

$$x = 1$$

$$f(1) + 2f'(1) + 2f'(1) + 2f(1) = 1$$

$$4f'(1) + 1 = 1 \Rightarrow f'(1) = 0$$

$$2xf'(x^2) + 4x^3 f''(x^2) + 4xf'(x^2) + 2xf'(x) + (x^2 + 1)f''(x) + 2f(x) + 2f'(x) = 0$$

$$x = 1$$

$$2f'(1) + 4f''(1) + 4f'(1) + 2f'(1) + 2f''(1) + 2f(1) + 2f'(1) = 0$$

$$6f''(1) + \frac{2}{3} = 0$$

$$f''(1) = -\frac{1}{9}$$



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PAPER-2

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

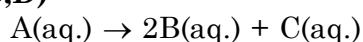
1. Ans. (A,B,C,D)

2. Ans. (A,B,C)

3. Ans. (A,D)

4. Ans. (A,D)

5. Ans. (A,B,D)



$$t = 0 \quad a \quad \quad \quad = 2$$

$$t = 15 \quad a - x \quad 2x \quad x = 3$$

$$t = \infty \quad a - a \quad 2a \quad a$$

$$\Rightarrow a + 2x = 3 \Rightarrow x = \frac{1}{2}$$

$$\Rightarrow K(15) = \ln \left(\frac{2}{3/2} \right) = 1.4 - 1.1$$

$$\Rightarrow K = \frac{0.3}{15} \text{ min}^{-1} = 0.02 \text{ min}^{-1}$$

$$\text{Rate} = k[A] = K \left(\frac{\pi}{RT} \right) = 0.02 \left(\frac{(3/2)}{(0.08 \times 300)} \right)$$

$$= 1.25 \times 10^{-3} \text{ M/min.}$$

6. Ans. (A, B)

$$[A]_0 = 0.8 \text{ M} ; [B]_0 = 0.002 \text{ M}$$

$$\therefore [A]_0 \gg [B]_0 \Rightarrow \text{rate} = K' [B]^2$$

$$K' = k[A]$$

$$t_{1/2} = \frac{1}{[B]_0 K'}$$

$$10 = \frac{1}{0.002 \cdot K(0.8)} \quad \dots\dots(1)$$

$$y = \frac{1}{0.004K(0.2)}$$

from (1) & (2)

$$\Rightarrow y = 20$$

$$x = \frac{1}{(0.002)K(0.4)} \quad \dots\dots(3)$$

from (1) & (3)

$$x = 20$$

$$\Rightarrow [A]_0 = 0.004 \text{ M} ; [B]_0 = 0.9 \Rightarrow \text{rate} = K''[A]$$

$$\Rightarrow t_{1/2} = \frac{\ln 2}{K''} = \frac{\ln 2}{K''}$$

$$20 = \frac{\ln 2}{K(0.9)^2} \quad \dots\dots(1)$$

$$z = \frac{\ln 2}{K(0.45)^2} \quad \dots\dots(2)$$

$$\text{Divide (1) \& (2)} ; \frac{z}{20} = \left(\frac{0.9}{0.45} \right)^2 = 4$$

$$\Rightarrow z = 80$$

7. Ans. (A)

8. Ans. (C)

9. Ans. (A)

10. Ans. (A)

SECTION-II

1. Ans. (3.00)

2. Ans. (6.00)

3. Ans. (9.00)

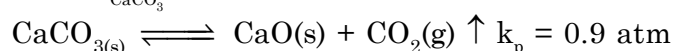
4. Ans. (3.00)

5. Ans. (6.00)

6. Ans. (4.00)

7. Ans. (80.00)

$$n_{\text{CaCO}_3} = 0.25$$



$$k_p = 0.9 = p_{\text{CO}_2}$$

$$\Rightarrow n_{\text{CO}_2} = \frac{0.9 \times 16.42}{0.0821 \times 900} = 0.2$$

$$\Rightarrow \% \text{ decomposition} = \frac{0.2}{0.25} \times 100 = 80\%$$

8. Ans. (6.25)

$$p_{\text{O}_2} = K_H \cdot \frac{n_{\text{O}_2}}{n_{\text{O}_2} + n_{\text{H}_2\text{O}}} \approx K_H \times \frac{n_{\text{O}_2}}{n_{\text{H}_2\text{O}}}$$

$$4 = K_H \times \frac{4 \times 12.5}{\frac{R \times 298}{200}} \quad \dots\dots(1)$$

$$2 = K_H \times \frac{2 \times V}{\frac{R \times 298}{100}} \quad \dots\dots(2)$$

$$\text{from (1) \& (2)} ;$$

$$V = 6.25 \text{ ml}$$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A,B,C,D)**

Sol. All function are differentiable

2. **Ans. (B,C)**

Sol. Apply continuity at $x = 1$,

$$2a + b = 2$$

Apply continuity at $x = 3$,

$$2a + 3b = 0$$

$$\text{from (1) \& (2) } a = \frac{3}{2} \text{ and } b = -1$$

3. **Ans. (D)**

$$\text{Sol. } 9\alpha^2 - 11\alpha + 1 = 0 \Rightarrow 9\alpha - 11 = -\frac{1}{\alpha}$$

$$\& 9\beta^2 - 11\beta + 1 = 0 \Rightarrow 11\beta - 1 = 9\beta^2$$

$$\therefore \frac{1}{(9\alpha - 11)^2} + \left(\frac{11\beta - 1}{9}\right) = \alpha^2 + \beta^2$$

$$= (\alpha + \beta)^2 - 2\alpha\beta = \frac{103}{81}$$

4. **Ans. (C)**

$$\text{Sol. } (\theta_1 + \theta_2) \left(\frac{\cos^4 x}{\theta_1} + \frac{\sin^4 x}{\theta_2} \right) = 1$$

$$\cos^4 x + \sin^4 x + \frac{\theta_2}{\theta_1} \cos^4 x + \frac{\theta_1}{\theta_2} \sin^4 x = \theta$$

$$1 - 2\sin^2 x \cos^2 x + \frac{\theta_2}{\theta_1} \cos^4 x + \frac{\theta_1}{\theta_2} \sin^4 x = 0$$

$$\left(\sqrt{\frac{\theta_2}{\theta_1}} \cos^2 x - \sqrt{\frac{\theta_1}{\theta_2}} \sin^2 x \right)^2 = 0$$

$$\tan^2 x = \frac{\theta_2}{\theta_1}$$

$$\theta_2 = \theta_1 \tan^2 x$$

$$\frac{d\theta_2}{d\theta_1} = \tan^2 x$$

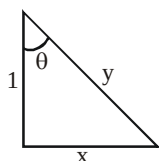
5. **Ans. (A,B,C,D)**

Sol. $y \sin \theta = x$

$$y \cos \theta = 1$$

$$(y^n - x^n) = \frac{1 - \sin^n \theta}{\cos^n \theta}$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin \theta}{\cos \theta} = 0, \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin^2 \theta}{\cos^2 \theta} = 1$$



$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin^3 \theta}{\cos^3 \theta} = \infty \Rightarrow \text{Limit does not exist.}$$

6. **Ans. (B)**

$$\text{Sol. } \lim_{h \rightarrow 0} \frac{(\log(2+h) - \log 2)3 \cdot (4^h - 1 - h)}{((8+h)^{1/3} - (4+3h)^{1/2}) \cdot \sin \pi(1+h)}$$

$$\lim_{h \rightarrow 0} \frac{3 \cdot \log \left(1 + \frac{h}{2} \right) \cdot (4^h - 1 - h)}{2 \left[\left(1 + \frac{h}{8} \right)^{1/3} - \left(1 + \frac{3h}{4} \right)^{1/2} \right] (-\sin \pi h)}$$

$$\lim_{h \rightarrow 0} \frac{3 \cdot \left(\log \left(\frac{1 + \frac{h}{2}}{\frac{h}{2}} \right) \cdot \frac{h}{2} \right) \cdot (4^h - 1 - h)}{2 \left[\left(1 + \frac{h}{24} + \dots \right) - \left(1 + \frac{3h}{8} \right) \right] \left(\frac{-\sin \pi h}{\pi h} \right) \pi h}$$

Using standard limit

$$\lim_{h \rightarrow 0} \frac{\left(\frac{3h}{2} \right) (4^h - 1 - h)}{2 \left(\frac{h}{3} \right) (\pi h)} = \lim_{h \rightarrow 0} \frac{9}{4\pi} \left(\frac{4^h - 1 - h}{h} \right)$$

Using LH rule

$$\lim_{h \rightarrow 0} \left(\frac{9}{4\pi} \right) \left(\frac{4^h \log 4 - 1}{1} \right)$$

$$\text{Taking limit } \left(\frac{9}{4\pi} \right) (\log 4 - 1)$$

7. **Ans. (B)**

$$\text{Sol. } f(x) = x^3 - 6x^2 - 36x + 2$$

$$f'(x) = 3x^2 - 12x - 36 = 3(x + 2)(x - 6)$$

$$f''(x) = 6x - 12 = 6(x - 2)$$

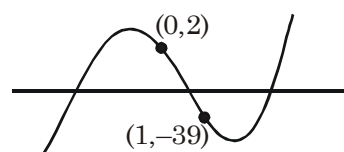
$$(P) \quad f'(x) > 0 \therefore x \in (-\infty, -2) \cup (6, \infty)$$

$$(Q) \quad f''(x) = 0 \therefore x = 2 \Rightarrow c = 2$$

$$(R) \quad f(x) = 0 \Rightarrow x^3 - 6x^2 - 36x + 2 = 0$$

$$\therefore f(0) = 2 \text{ and } f(1) = -39$$

$\Rightarrow$ three real roots



$$(S) \quad \text{for rolles theorem } f(a) = f(0) \\ a^3 - 6a^2 - 36a + 2 = 2$$

$$\therefore a = 0, \frac{6 + \sqrt{180}}{2}, \frac{6 - \sqrt{180}}{2}$$

$$\therefore a > 2 \Rightarrow k = \frac{6 + \sqrt{180}}{2}$$

8. Ans. (C)

Sol. (P) $y = -2\ln|x|$ as $|x| < 1$

$$\Rightarrow y' = -\frac{2}{x} \Rightarrow y'\left(-\frac{1}{2}\right) = 4$$

$$(Q) \quad f(1) = \frac{1}{2} \Rightarrow g\left(\frac{1}{2}\right) = 1$$

$$g''(y) = -\frac{f''(x)}{(f'(x))^3}$$

$$g''\left(\frac{1}{2}\right) = -\frac{f''(1)}{(f'(1))^3}$$

$$\text{also } f' = \frac{3}{2}x^2 + \frac{1}{1+x^2}$$

$$f'' = 3x - \frac{2x}{(1+x^2)^2}$$

$$f'(1) = 2, f''(1) = 3 - \frac{1}{2} = \frac{5}{2}$$

$$g''\left(\frac{1}{2}\right) = -\frac{\frac{5}{2}}{(2)^3} = -\frac{5}{16}$$

$$(R) \quad \ell_{ny} = x^x \ell_{nx}$$

$$\frac{1}{y}y' = x^x(1 + \ell_{nx})\ell_{nx} + \frac{x^x}{x} \Rightarrow y'(1) = 1$$

$$\Rightarrow \frac{yy'' - (y')^2}{y^2}$$

$$= x^x(1 + \ell_{nx})^2 \ell_{nx} + x^x \left(\frac{1}{x} + \frac{2\ell_{nx}}{x} \right) + \frac{x^x(1 + \ell_{nx})}{x} - \frac{x^x}{x^2}$$

$$\text{at } x = 1 \Rightarrow y'' - 1 = 0 + 1 + 1 - 1 \Rightarrow y'' = 2$$

9. Ans. (A)

Sol. (P) $f'(x) = 0$ must have two real and unequal roots

$$\Rightarrow f'(x) = 6x^2 - 6(2 + \lambda)x + 12\lambda$$

$$D > 0 \Rightarrow (2 + \lambda)^2 - 8\lambda > 0 \Rightarrow (\lambda - 2)^2 > 0$$

(Q) Take log & differentiable

(R) Point of intersection is $x = 1$
now proceed

$$(S) \quad \frac{dy}{dx} = 0 \text{ or } \infty$$

10. Ans. (D)

Sol. Case-I : If $a > 0$, then

$$g \circ f(x) = \begin{cases} x + a + 1; & x < -a \\ (x + a)^2 + 2(b - x - a) + 1; & -a \leq x < 0 \\ (1 - x)^2 + 2(b - 1 + x) + 1; & 0 \leq x < 1 \\ (x - 1)^2 + 2(b - x + 1) + 1; & x \geq 1 \end{cases}$$

$$a = 1 \text{ \& } b = 0$$

Case-II : If $a = 0$, then

$$g(f(x)) = \begin{cases} x + 1 & x < 0 \\ x^2 + 2b & 0 \leq x < 1 \\ (x - 2)^2 + 2b & x \geq 1 \end{cases}$$

$$a = 0; b = \frac{1}{2}$$

$$\left. \frac{d(g(f(x)))}{dx} \right|_{x=4} = (2(x - 1) - 2)|_{x=4} = 4$$

SECTION-II

1. Ans. 0.75

$$\text{Sol. } \therefore 2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$\Rightarrow 2.4 \frac{dx}{dt} + 2.3(-1) = 0 \Rightarrow \frac{dx}{dt} = \frac{3}{4}$$

2. Ans. 4.00

$$\text{Sol. Required limit is } \left. \frac{d}{dx}(h^{-1}(x)) \right|_{x=1}$$

$$\therefore f^{-1}(x) = x - 1; g^{-1}(x) = x^3 + x + 1$$

$$\text{Also } h(x) = g(f(x))$$

$$\Rightarrow h^{-1}(x) = f^{-1}(g^{-1}(x)) = x^3 + x$$

$$\Rightarrow \frac{d}{dx}(h^{-1}(x)) = 3x^2 + 1$$

3. Ans. 3.57

$$\text{Sol. } \therefore y = \frac{2x^2 + 7x + 3}{2x^2 + x + 1} \quad \dots(i)$$

$$\Rightarrow 2x^2y + xy + y = 2x^2 + 7x + 3$$

$$\Rightarrow 2x^2(y - 1) + x(y - 7) + (y - 3) = 0$$

$$\text{if } y - 1 \neq 0$$

$$\therefore x \in \mathbb{R} \Rightarrow D \geq 0$$

$$(y - 7)^2 - 8(y - 1)(y - 3) \geq 0$$

$$\Rightarrow y^2 - 14y + 49 - 8y^2 + 32y - 24 \geq 0$$

$$\Rightarrow -7y^2 + 18y + 25 \geq 0$$

$$\Rightarrow 7y^2 - 18y - 25 \leq 0$$

$$\Rightarrow (y + 1)(7y - 25) \leq 0$$

$$-1 \leq y \leq \frac{25}{7} \quad \text{but } y \neq 1$$

put $y = 1$ in (1) $\therefore$ (1)

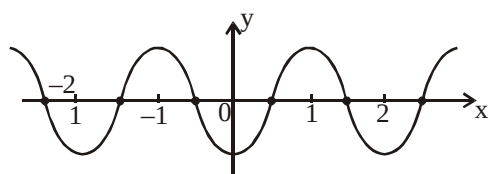
$$\Rightarrow 2x^2 + 7x + 3 = 2x^2 + x + 1$$

$$\Rightarrow x = -\frac{1}{3}$$

$$\therefore \text{Final range} = \left[-1, \frac{25}{7}\right]$$

4. Ans. 6.00

Sol. $\lim_{x \rightarrow \infty} P(x) \rightarrow \infty$ and $\lim_{x \rightarrow (-\infty)} P(x) \rightarrow \infty$.



$P(0) = -1$ and $P(x)$ is even this $P(1) = P(-1) = 1$ & $P(2) = P(-2) = -1$.

Hence, from IVT minimum no. of zeroes are 6.

5. Ans. 8.00

Sol. $\therefore$ For integral roots of equation

$$ax^2 + bx + c = 0$$

$$a = 1; b, c \in I$$

and $D = b^2 - 4c$ should be a perfect square

$\therefore$ for equation $4 + 4n$ should be a perfect square

$\Rightarrow (n + 1)$ should be perfect square

$$\therefore n + 1 = 9, 16, 25, 36, 49, 64, 81, 100$$

$$\therefore n = 8, 15, 24, 35, 48, 63, 80, 99$$

6. Ans. 1.00

Sol. Using L'Hospital :

$$L = \left| \lim_{x \rightarrow \frac{3\pi}{4}} \frac{4(2 \sin x \cos^2 x - \sin^3 x) + \sin x + \cos x}{\cos x - \sin x} \right|$$

$$= |-1| = 1$$

7. Ans. 1.00

$$\text{Sol. } f'(0^+) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(h^2 + h^3) - h}{h} + 1$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(h^2 + h^3) (h^2 + h^3)}{h^2 (h^2 + h^3)} = 1$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} \frac{\sin^{-1}(h^2 + h^3) - h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{\sqrt{1 - (h^2 + h^3)^2}} \cdot (2h + 3h^2) - 1}{1} = -1$$

8. Ans. 1.00

$$\text{Sol. } f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5} \quad (f(0) = 0)$$

$$f'(x) = 1 + x + x^2 + x^3 + x^4 \quad (f'(0) = 1)$$

$$f''(x) = 1 + 2x + 3x^2 + 4x^3 \quad (f''(0) = 1)$$

Now $f(g(x)) = x$

Diff. w.r.to x

$$f'(g(x))g'(x) = 1 \quad \dots(i)$$

again Diff. w.r.to x

$$f''(g(x))(g'(x))^2 + g''(x)f'(g(x)) = 0$$

$$x = 0$$

$$f''(g(0))(g'(0))^2 + g''(0)f'(g(0)) = 0 \quad \dots(ii)$$

$$\text{as } f(0) = 0 \Rightarrow g(0) = 0$$

$$\text{and from (i) } g'(0) = \frac{1}{f'(0)}$$

now from (ii)

$$f''(0) \left(\frac{1}{f'(0)} \right)^2 + g''(0)f'(0) = 0$$

$$1.1^2 + g''(0).1 = 0$$

$$g''(0) = -1$$

$$|g''(0)| = 1$$